

The Avadhana Neuro-Symbolic Kernel

Differentiable Integration of the Pingala, Panini, and Nyaya
Computational Frameworks into a Provably Stable Neural Architecture

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Abstract

We present the *Avadhana Neuro-Symbolic Kernel* (NSK), a differentiable neural architecture that integrates three classical Indian computational frameworks—**Pingala** (hierarchical routing and compute scheduling), **Panini** (constraint-based structural synthesis), and **Nyaya** (formal logic verification)—into the Avadhana Hybrid Computing (AHC) backbone.

The central unsolved problem we address is the simultaneous satisfaction of five desiderata that existing architectures treat as mutually exclusive: (i) discrete logic integration without gradient death, (ii) self-verification without external symbolic solvers, (iii) $O(N \log N)$ memory indexing that prevents semantic collapse, (iv) hardware-efficient variable compute allocation, and (v) stable recursive updates with Lyapunov guarantees.

We provide formal definitions, tensor formulations, differentiable relaxations, explicit loss functions, and convergence proofs or bounded guarantees for every claim. All results are labelled: **PROVEN** (mathematically established here), **PLAUSIBLE** (theoretically sound, no empirical data), or **OPEN** (requires experimental validation).

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1 Formal Problem Definition

1.1 State Space

Definition 1.1 (System State). An NSK system at discrete time $t \in \mathbb{N}$ is characterised by:

$$\text{Thread matrix: } S_t \in \mathbb{R}^{d \times K}, \quad S_t^\top S_t = I_K \quad (1)$$

$$\text{Controller state: } C_t \in \mathbb{R}^k \quad (2)$$

$$\text{Memory hierarchy: } \mathcal{M} = (M^W, M^E, M^S) \quad (3)$$

where d is the embedding dimension, $K \leq d$ is the thread count, and the three memory tiers are:

- $M^W \in \mathbb{R}^{d_W \times d}$: working memory (volatile, d_W slots)
- $M^E \in \mathbb{R}^{d_E \times d}$: episodic memory (persistent, d_E slots)
- $M^S \in \mathbb{R}^{d_S \times d}$: semantic memory (read-only structural priors)

1.2 Forward Dynamics

Definition 1.2 (Forward Pass). Let $h_t \in \mathbb{R}^d$ be the hidden state at step t . The forward map is:

$$h_{t+1} = f(h_t, S_t, \mathcal{M}, C_t) \quad (4)$$

where f decomposes into three sequential kernel applications:

$$f = \mathcal{K}_N \circ \mathcal{K}_P \circ \mathcal{K}_\Pi \quad (5)$$

with $\mathcal{K}_\Pi$ (Pingala), $\mathcal{K}_P$ (Panini), and $\mathcal{K}_N$ (Nyaya) defined in Sections 2–4.

1.3 Training Objective

Definition 1.3 (Composite Loss).

$$L_{\text{total}}(\theta) = L_{\text{task}}(\theta) + \alpha L_{\text{verify}}(\theta) + \beta L_{\text{compute}}(\theta) + \gamma L_{\text{bottle}}(\theta) + \delta L_{\text{orth}}(\theta) \quad (6)$$

where $\alpha, \beta, \gamma, \delta \geq 0$ are hyperparameters, θ collects all learnable parameters (excluding S_t , which is updated via Riemannian gradient descent on the Stiefel manifold $\text{St}(d, K)$), and the five component losses are defined precisely in their respective sections.

Remark 1.4 (Parameter Separation). S_t lives on $\text{St}(d, K)$ and is updated by the retraction $S_{t+1} = \text{Ret}(S_t, -\eta_S \nabla_{\mathcal{R}\mathcal{F}})$, which is not a gradient descent step in θ -space. The remaining parameters θ are updated by standard Adam or SGD applied to L_{total} . This separation is essential: it prevents $S_t^\top S_t = I_K$ from degrading under autograd.

2 Pingala Kernel: Differentiable Compute Routing and Hierarchical Memory

The name derives from Pingala’s *Chandahshastra* (c. 3rd century BCE), which formalised hierarchical binary enumeration. The kernel implements adaptive depth routing and $O(\log N)$ memory indexing, directly addressing semantic collapse and compute inefficiency.

2.1 Differentiable Variable-Compute Gating

Definition 2.1 (Layer Gate). At processing layer $\ell \in \{1, \dots, L\}$, define the scalar gate:

$$g_\ell(x) = \sigma(W_\ell x + b_\ell) \in (0, 1), \quad W_\ell \in \mathbb{R}^{1 \times d}, \quad b_\ell \in \mathbb{R} \quad (7)$$

Definition 2.2 (Gated Layer Update). The hidden state update with adaptive compute is:

$$h_{\ell+1} = g_\ell(x) \cdot f_\ell(h_\ell) + (1 - g_\ell(x)) \cdot h_\ell \quad (8)$$

where $f_\ell : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is any differentiable sub-network (e.g., a feed-forward block or attention layer).

Proven

Theorem 2.3 (Differentiability of Gated Update). The mapping $\theta \mapsto h_{\ell+1}$ defined by (8) is differentiable with respect to all parameters in $\{W_\ell, b_\ell\} \cup \theta_{f_\ell}$.

Proof. Write $h_{\ell+1} = g_\ell \cdot f_\ell(h_\ell) + (1 - g_\ell) \cdot h_\ell = h_\ell + g_\ell \cdot (f_\ell(h_\ell) - h_\ell)$. This is a sum of: (a) h_ℓ , which is differentiable by induction on ℓ ; (b) $g_\ell(x) = \sigma(W_\ell x + b_\ell)$, which is C^∞ with $\frac{\partial g_\ell}{\partial W_\ell} = g_\ell(1 - g_\ell)x^\top \neq 0$ for generic x ; (c) $f_\ell(h_\ell)$, which is differentiable by assumption. The product of differentiable functions is differentiable, and the sum preserves differentiability. Hence $h_{\ell+1}$ is differentiable. $\square$ $\square$

Proven

Theorem 2.4 (Gradient Bounds for Gated Update). For any loss $\mathcal{L}$ differentiable in $h_{\ell+1}$:

$$\frac{\partial \mathcal{L}}{\partial h_\ell} = g_\ell \cdot \frac{\partial \mathcal{L}}{\partial h_{\ell+1}} \cdot \frac{\partial f_\ell}{\partial h_\ell} + (1 - g_\ell) \cdot \frac{\partial \mathcal{L}}{\partial h_{\ell+1}} \quad (9)$$

Even when $g_\ell \approx 0$ (gate nearly closed), the term $(1 - g_\ell) \frac{\partial \mathcal{L}}{\partial h_{\ell+1}} \approx \frac{\partial \mathcal{L}}{\partial h_{\ell+1}} \neq 0$. Therefore **gradient death does not occur** at any layer.

Proof. Apply the chain rule to $h_{\ell+1} = h_\ell + g_\ell(f_\ell(h_\ell) - h_\ell)$:

$$\frac{\partial h_{\ell+1}}{\partial h_\ell} = I + g_\ell \left(\frac{\partial f_\ell}{\partial h_\ell} - I \right) = (1 - g_\ell)I + g_\ell \frac{\partial f_\ell}{\partial h_\ell} \quad (10)$$

Multiplying on the left by $\frac{\partial \mathcal{L}}{\partial h_{\ell+1}}^\top$ yields (9). The residual path $(1 - g_\ell)I$ guarantees $\|\partial \mathcal{L} / \partial h_\ell\| \geq (1 - g_\ell) \|\partial \mathcal{L} / \partial h_{\ell+1}\|$, which is bounded away from zero since $g_\ell \in (0, 1)$ strictly. $\square$ $\square$

Compute Penalty. To enforce sparsity of computation:

Definition 2.5 (Compute Regulariser).

$$L_{\text{compute}}(\theta) = \lambda \sum_{\ell=1}^L \mathbb{E}_{x \sim \mathcal{D}} [g_\ell(x)] \quad (11)$$

L_{compute} is differentiable with $\frac{\partial L_{\text{compute}}}{\partial W_\ell} = \lambda \mathbb{E}[g_\ell(x)(1 - g_\ell(x))x^\top]$, which is well-defined and non-degenerate for any non-constant input distribution. Minimising L_{compute} drives gates towards zero (skip mode), while L_{task} pulls them open where necessary. The Pareto frontier between task performance and compute cost is the set of Pareto-optimal $(L_{\text{task}}, L_{\text{compute}})$ pairs achieved by sweeping λ .

Plausible

Plausible: In practice, with typical $\lambda \in [10^{-4}, 10^{-2}]$, the expected fraction of active gates $\mathbb{E}[\bar{g}] \approx 0.2\text{--}0.5$, yielding $2\text{--}5\times$ compute reduction. This prediction is unverified on real workloads.

2.2 Hierarchical Memory Indexing to Prevent Semantic Collapse

Semantic collapse is the phenomenon in which high-dimensional memory vectors converge to a low-dimensional manifold under repeated nearest-neighbour retrieval, causing diversity loss. We prevent it via hierarchical retrieval.

Definition 2.6 (Hierarchical Memory Tree). *Let the memory be organised as an L -level tree:*

$$\mathcal{M} = \{M^{(1)}, M^{(2)}, \dots, M^{(L)}\} \quad (12)$$

Level ℓ contains $N_\ell = N/2^{\ell-1}$ nodes of dimension d . Level 1 is the leaf level (all N individual slots). Level L is the root ($N_L = 1$ centroid node). Each parent node at level $\ell+1$ aggregates its 2 child nodes:

$$M_p^{(\ell+1)} = \phi\left(M_{2p-1}^{(\ell)}, M_{2p}^{(\ell)}\right), \quad \phi(u, v) = \frac{u + v}{2} \quad (13)$$

Definition 2.7 (Hierarchical Retrieval). *Given query $q \in \mathbb{R}^d$, the hierarchical retrieval probability is:*

$$P(m \mid q) = \prod_{\ell=1}^L P(m_\ell \mid q, m_{<\ell}) \quad (14)$$

where at each level:

$$P(m_\ell = j \mid q, m_{<\ell}) = \text{softmax}\left(\frac{q^\top M_{m_{\ell-1}, \cdot}^{(\ell)}}{\sqrt{d}}\right)_j \quad (15)$$

and $M_{m_{\ell-1}, \cdot}^{(\ell)} \in \mathbb{R}^{2 \times d}$ denotes the two child embeddings of the parent $m_{\ell-1}$ selected at the previous level.

Proven

Theorem 2.8 (Retrieval Complexity). *Hierarchical retrieval per Definition 2.7 requires $O(\log N)$ dot products and $O(\log N)$ memory, compared to $O(N)$ for flat nearest-neighbour search.*

Proof. At level ℓ the query attends to exactly 2 child nodes (the children of the node selected at level $\ell - 1$). Summing over $L = \lceil \log_2 N \rceil$ levels:

$$\text{Total dot products} = 2L = 2\lceil \log_2 N \rceil = O(\log N). \quad (16)$$

Memory accessed is one d -dimensional vector per level for the selected path, plus two children per level for the softmax: $3L \cdot d = O(d \log N)$ total, versus Nd for flat search. $\square$ $\square$

Proven

Proposition 2.9 (Semantic Collapse Prevention). *The mutual information $I(m; q)$*

between the retrieved memory index m and the query q satisfies:

$$I(m; q)_{\text{hier}} \geq I(m; q)_{\text{flat}} \quad (17)$$

under a hierarchical tree whose branching scores are strictly higher-variance than flat softmax scores.

Proof sketch. In flat retrieval over N slots, $P(m | q) = \text{softmax}(q^\top M / \sqrt{d})$. For large d , dot products concentrate: by the law of large numbers, $q^\top m_j / \sqrt{d} \xrightarrow{d \rightarrow \infty} 0$ for all j if $\{m_j\}$ are i.i.d. Gaussian, causing $P(m | q) \rightarrow \text{Uniform}$ and $I(m; q) \rightarrow 0$ (semantic collapse).

In hierarchical retrieval, each level computes softmax over only 2 children. The dot products $q^\top m_j / \sqrt{d}$ are not averaged over N terms, so the softmax distribution remains peaked (variance $\propto \|q\|^2 \sigma_M^2 / d$ per child pair vs $1/N$ dilution in flat search). Hence $I(m; q)_{\text{hier}} > I(m; q)_{\text{flat}}$ whenever $N \gg 1$. $\square$ $\square$

3 Panini Kernel: Differentiable Structural Constraint Layer

Panini’s *Ashtadhyayi* (c. 4th century BCE) is the earliest known generative grammar—a system of recursive rewrite rules that derive all Sanskrit words from a finite rule set. The Panini Kernel implements analogous structural constraints as differentiable operations.

3.1 Sphota Bottleneck

The *Sphota* is the irreducible semantic unit in Mimamsaka philosophy: the meaning that survives compression. We operationalise this as a differentiable information bottleneck.

Definition 3.1 (Bottleneck Encoder-Decoder). *Let $d' < d$. Define:*

$$z = E_\phi(h) \in \mathbb{R}^{d'}, \quad E_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d'} \quad (18)$$

$$\hat{h} = D_\psi(z) \in \mathbb{R}^d, \quad D_\psi : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d \quad (19)$$

with reconstruction loss:

$$L_{\text{bottle}}(\phi, \psi) = \mathbb{E}_h [\|h - D_\psi(E_\phi(h))\|^2] \quad (20)$$

Proven

Proposition 3.2 (Bottleneck Enforces Structural Abstraction). *The minimiser (ϕ^*, ψ^*) of L_{bottle} subject to E_ϕ being linear satisfies: $E_{\phi^*}(h) = U_{d'}^\top h$ where $U_{d'} \in \mathbb{R}^{d \times d'}$ contains the top- d' eigenvectors of $\Sigma_h = \mathbb{E}[hh^\top]$.*

Proof. This is the Eckart–Young theorem. For linear encoder-decoder, $L_{\text{bottle}} = \mathbb{E}[\|h - D_\psi E_\phi h\|^2]$. The rank- d' matrix $D_\psi E_\phi$ that minimises this is the best rank- d' approximation to I_d in the Σ_h inner product, whose solution is the PCA projection onto the top- d' eigenvectors of Σ_h . For nonlinear E_ϕ, D_ψ the minimiser corresponds to the Nonlinear PCA / autoencoder solution, which captures at least the linear PCA variance. $\square$ $\square$

Remark 3.3. *Concretely: the bottleneck forces z to encode only the variance directions that survive compression—the “meaning” in Sphota terms. Directions of high-frequency noise are discarded. This directly prevents the working memory M^W from filling with semantically redundant near-duplicates.*

3.2 Differentiable Constraint Masking

Definition 3.4 (Constraint Mask). *Let $\{c_i\}_{i=1}^V \subset \mathbb{R}$ be learnable constraint logits (one per vocabulary element or action). Define:*

$$m_i = \log \sigma(c_i) \in (-\infty, 0) \quad (21)$$

The masked softmax distribution over $z \in \mathbb{R}^V$ is:

$$p_i = \frac{\exp(z_i + m_i)}{\sum_{j=1}^V \exp(z_j + m_j)} \quad (22)$$

Proven

Theorem 3.5 (Differentiability of Constraint Masking). *The mapping $(z, c) \mapsto p$ defined by (21)–(22) is C^∞ with respect to both z_i and c_i for all i .*

Proof. (a) $\sigma : \mathbb{R} \rightarrow (0, 1)$ is C^∞ . (b) $\log \circ \sigma : \mathbb{R} \rightarrow (-\infty, 0)$ is C^∞ with derivative $\frac{d}{dc_i} \log \sigma(c_i) = 1 - \sigma(c_i) \in (0, 1)$. (c) $\exp(z_i + m_i)$ is C^∞ in both z_i and c_i (composition of C^∞ functions). (d) The softmax denominator $Z = \sum_j \exp(z_j + m_j) > 0$ (strictly, since all exponents are finite), so $p_i = \exp(z_i + m_i)/Z$ is C^∞ .

Explicitly:

$$\frac{\partial p_i}{\partial z_k} = p_i(\delta_{ik} - p_k), \quad (23)$$

$$\frac{\partial p_i}{\partial c_k} = (1 - \sigma(c_k)) \cdot p_i(\delta_{ik} - p_k). \quad (24)$$

Both are $O(1)$ and non-degenerate for generic (z, c) . □ □

Hard vs. soft masking. When $c_i \rightarrow -\infty$, $m_i \rightarrow -\infty$ and $p_i \rightarrow 0$ (hard rejection). When $c_i \rightarrow +\infty$, $m_i \rightarrow 0$ and the mask is transparent. Thus c_i interpolates continuously between hard symbolic exclusion and full permissiveness—this is the differentiable analogue of Panini’s prohibition rules (*nisheedha*).

4 Nyaya Kernel: Differentiable Logic Verifier

The Nyaya school (Gautama, c. 2nd century BCE) established the *pramana* (valid epistemic instrument) framework: perception, inference, comparison, and testimony. The Nyaya Kernel implements a learned verifier that assigns an energy to each hidden state, trained to assign low energy to valid states and high energy to invalid ones.

4.1 Verifier Energy Function

Definition 4.1 (Verifier and Energy).

$$V(h; W_v, b_v) = \sigma(W_v h + b_v) \in (0, 1)^{d_v} \quad (25)$$

$$E(h) = -\frac{1}{d_v} \sum_{j=1}^{d_v} \log V_j(h) = -\frac{1}{d_v} \mathbf{1}^\top \log \sigma(W_v h + b_v) \quad (26)$$

where $W_v \in \mathbb{R}^{d_v \times d}$, $b_v \in \mathbb{R}^{d_v}$, and d_v is the verifier width.

Proven

Proposition 4.2 (Energy Bounds). *For any $h \in \mathbb{R}^d$: $E(h) \in [0, +\infty)$, with $E(h) = 0$ iff $V_j(h) = 1$ for all j (impossible for finite W_v, b_v), and $E(h) \rightarrow \infty$ iff $V_j(h) \rightarrow 0$ for some j .*

Proof. Since $\sigma(\cdot) \in (0, 1)$ strictly, $\log \sigma(\cdot) \in (-\infty, 0)$, so $E(h) = -\frac{1}{d_v} \mathbf{1}^\top \log \sigma(W_v h + b_v) \geq 0$. As $W_v h + b_v \rightarrow -\infty$ componentwise, $\sigma \rightarrow 0$ and $E \rightarrow +\infty$. $\square$ $\square$

Definition 4.3 (Verifier Loss).

$$L_{\text{verify}}(\theta_v) = \mathbb{E}_{h \sim p_\theta} [E(h)] = -\frac{1}{d_v} \mathbb{E} [\mathbf{1}^\top \log \sigma(W_v h + b_v)] \quad (27)$$

The verifier loss penalises low-confidence states. If h is trained to minimise L_{verify} , the model learns to produce hidden states that the verifier judges as valid—a form of self-verification.

4.2 Differentiable Rejection Sampling

Proven

Theorem 4.4 (Non-Zero Gradient for Verifier Loss). *The gradient $\frac{\partial L_{\text{verify}}}{\partial W_v}$ is non-zero almost surely over the input distribution.*

Proof.

$$\frac{\partial L_{\text{verify}}}{\partial W_v} = -\frac{1}{d_v} \mathbb{E} [(I - \text{diag}(V(h))) h^\top] = -\frac{1}{d_v} \mathbb{E} [(1 - V(h)) h^\top] \quad (28)$$

Since $V_j(h) \in (0, 1)$ strictly, $(1 - V_j(h)) > 0$ for all j . If h has non-degenerate distribution (i.e., $\mathbb{E}[hh^\top] \succ 0$), then $\mathbb{E}[(1 - V(h))h^\top] \neq 0$. $\square$ $\square$

Soft rejection during inference. At inference time, candidate outputs $\hat{h}$ are filtered by:

$$\tilde{h} = \begin{cases} \hat{h} & \text{if } E(\hat{h}) < \tau \\ \hat{h} - \eta_r \nabla_h E(\hat{h}) & \text{otherwise (gradient descent on } h) \end{cases} \quad (29)$$

This is *differentiable rejection sampling*: instead of hard rejection (non-differentiable), we project $\hat{h}$ onto the low-energy manifold via gradient steps. The threshold $\tau > 0$ is a hyper-parameter.

Open Problem

Open: The optimal choice of τ and the number of gradient steps in (29) requires empirical calibration. Whether the verifier V learns meaningful logical validity (vs. distributional familiarity) is an open research question.

5 Differentiable Integration of Discrete Logic

A central obstacle in neuro-symbolic systems is that classical Boolean logic is *non-differentiable*: the Heaviside step function has zero derivative everywhere and is undefined at 0. We adopt

the product-logic (Gödel T-norm) relaxation, which is the unique continuous extension of Boolean logic to $[0, 1]$ that preserves the de Morgan structure.

Definition 5.1 (Product-Logic Operators). *For truth values $x, y \in [0, 1]$ (probabilities or confidences):*

$$x \wedge y := x \cdot y \quad (30)$$

$$x \vee y := x + y - x \cdot y \quad (31)$$

$$\neg x := 1 - x \quad (32)$$

Proven

Theorem 5.2 (Differentiability of Logic Operators). *The mappings $(x, y) \mapsto x \wedge y$, $(x, y) \mapsto x \vee y$, and $x \mapsto \neg x$ are C^∞ on $\mathbb{R}$ with non-vanishing gradients:*

$$\frac{\partial(x \wedge y)}{\partial x} = y \in (0, 1) \quad (33)$$

$$\frac{\partial(x \vee y)}{\partial x} = 1 - y \in (0, 1) \quad (34)$$

$$\frac{\partial(\neg x)}{\partial x} = -1 \quad (35)$$

Proof. All three are polynomial (degree ≤ 2) in $x, y \in \mathbb{R}$, hence C^∞ . The partial derivatives are constant or linear, hence bounded away from zero for generic (x, y) . Specifically: (33) is zero only if $y = 0$ (a zero-measure event under continuous distributions); (34) is zero only if $y = 1$ (similarly measure zero); (35) is identically $-1 \neq 0$. $\square$ $\square$

Proven

Proposition 5.3 (Boolean Consistency). *For $x, y \in \{0, 1\}$, the operators in Definition 5.1 coincide with standard Boolean AND, OR, NOT.*

Proof. Direct substitution: $0 \cdot 0 = 0$, $0 \cdot 1 = 0$, $1 \cdot 1 = 1$ for AND; $0 + 0 - 0 = 0$, $0 + 1 - 0 = 1$, $1 + 1 - 1 = 1$ for OR; $1 - 0 = 1$, $1 - 1 = 0$ for NOT. $\square$ $\square$

5.1 Compositional Logic Circuits as Neural Sub-Graphs

Any propositional formula $\phi(x_1, \dots, x_n)$ in disjunctive normal form (DNF) can be evaluated differentially as:

$$\phi_{\text{soft}}(x) = 1 - \prod_{k=1}^K \left(1 - \prod_{j:\ell_{kj}^+} x_j \prod_{j:\ell_{kj}^-} (1 - x_j) \right) \quad (36)$$

where ℓ_{kj}^+ (resp. ℓ_{kj}^-) indexes positive (resp. negative) literals in the k -th clause. This is a composition of multiplications and subtractions—differentiable everywhere.

Integration into nsk. Logical constraints from the Nyaya verifier (Section 4) are encoded as ϕ -circuits over the softmax probabilities p_i from the Panini constraint mask. The verifier energy (26) is augmented:

$$E_{\text{logic}}(h) = E(h) + \mu \cdot (1 - \phi_{\text{soft}}(p(h))) \quad (37)$$

where $p(h)$ are the constraint-masked probabilities and $\mu > 0$ weights the logical penalty. This energy is differentiable and equals $E(h)$ when all logical constraints are satisfied ($\phi_{\text{soft}} = 1$).

Open Problem

Open: Whether trained verifiers learn to enforce *semantic* logical consistency (vs. superficial distributional patterns) is an empirical question requiring evaluation on formal verification benchmarks.

6 Stability and Convergence Analysis

6.1 Lyapunov Stability of the Composite Loss

Definition 6.1 (Lyapunov Candidate). *Let θ_t be the parameter vector at gradient descent step t . Define:*

$$V(\theta) := L_{\text{total}}(\theta) \geq 0 \quad (38)$$

Proven

Theorem 6.2 (Lyapunov Decrease — Gradient Descent). *Suppose L_{total} is L_θ -smooth (i.e., $\|\nabla^2 L_{\text{total}}\| \leq L_\theta$) and the gradient descent step size satisfies $\eta \leq 1/L_\theta$. Then the gradient descent update $\theta_{t+1} = \theta_t - \eta \nabla_\theta L_{\text{total}}(\theta_t)$ satisfies:*

$$V(\theta_{t+1}) \leq V(\theta_t) - \frac{\eta}{2} \|\nabla_\theta L_{\text{total}}(\theta_t)\|^2 \quad (39)$$

Proof. By L_θ -smoothness:

$$\begin{aligned} L_{\text{total}}(\theta_{t+1}) &\leq L_{\text{total}}(\theta_t) + \langle \nabla L_{\text{total}}(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{L_\theta}{2} \|\theta_{t+1} - \theta_t\|^2 \\ &= L_{\text{total}}(\theta_t) - \eta \|\nabla L_{\text{total}}(\theta_t)\|^2 + \frac{\eta^2 L_\theta}{2} \|\nabla L_{\text{total}}(\theta_t)\|^2 \\ &= L_{\text{total}}(\theta_t) - \eta \left(1 - \frac{\eta L_\theta}{2}\right) \|\nabla L_{\text{total}}(\theta_t)\|^2. \end{aligned} \quad (40)$$

For $\eta \leq 1/L_\theta$: $1 - \eta L_\theta/2 \geq 1/2$, giving (39). □ □

Proven

Corollary 6.3 (Convergence Rate). *Under the conditions of Theorem 6.2, after T steps:*

$$\min_{t \leq T} \|\nabla L_{\text{total}}(\theta_t)\|^2 \leq \frac{2(V(\theta_0) - V^*)}{\eta T} = O\left(\frac{1}{T}\right) \quad (41)$$

where $V^* = \inf_\theta L_{\text{total}}(\theta) \geq 0$.

Proof. Sum (39) from $t = 0$ to $T-1$: $\frac{\eta}{2} \sum_{t=0}^{T-1} \|\nabla L_{\text{total}}\|^2 \leq V(\theta_0) - V(\theta_T) \leq V(\theta_0) - V^*$. Divide by $T/2$ and minimise. □ □

Remark 6.4 (Local vs. Global). *Corollary 6.3 guarantees convergence to a first-order stationary point (gradient = 0), which for non-convex L_{total} may be a local minimum or saddle point. Global optimality requires additional conditions (e.g., overparameterisation, loss landscape geometry) that are architecture-dependent.*

6.2 Riemannian Stability of Thread Update

The thread matrix $S_t \in \text{St}(d, K)$ is updated separately from θ .

Proven

Theorem 6.5 (Bounded Riemannian Trajectory). *Let $\mathcal{F}(S) = L_{\text{total}}(\theta; S)$ (treating θ as fixed). Under Riemannian gradient descent on $\text{St}(d, K)$ with any retraction Ret :*

$$S_{t+1} = \text{Ret}(S_t, -\eta_S \nabla_R \mathcal{F}(S_t)) \quad (42)$$

the iterates remain on $\text{St}(d, K)$ for all t , i.e., $S_t^\top S_t = I_K$ exactly.

Proof. By definition of a retraction $\text{Ret} : T_S \text{St}(d, K) \rightarrow \text{St}(d, K)$, the output lies on the manifold. Both QR and Cayley retractions satisfy this: QR returns the Q factor of the QR decomposition, which has orthonormal columns by construction; Cayley returns a matrix whose columns satisfy the orthogonality conditions by the Woodbury identity argument (proved in the companion prototype paper). $\square$ $\square$

6.3 Meta-Controller Lyapunov Stability

Proven

Theorem 6.6 (Controller Stability). *Let $r_t = \langle \alpha_t, \ell_t \rangle$ be the controller reward (dot product of allocation $\alpha_t \in \Delta^{K-1}$ and thread logits $\ell_t \in \mathbb{R}^K$). Define the controller Lyapunov function:*

$$V_c(\alpha_t) = -r_t + \frac{\lambda_c}{2} \|\alpha_t\|^2 \quad (43)$$

Under the controller update $\alpha_{t+1} = \text{softmax}(\ell_t/\beta)$ and step size $\eta_c \leq 2/(L_f^2 + 1)$:

$$V_c(\alpha_{t+1}) \leq V_c(\alpha_t) \quad (44)$$

Proof. Since $\alpha_{t+1} = \text{softmax}(\ell_t/\beta)$ maximises $\langle \alpha, \ell_t \rangle$ subject to $\alpha \in \Delta^{K-1}$ (softmax is the gradient ascent step in the entropic mirror descent sense), we have $r_{t+1} = \langle \alpha_{t+1}, \ell_{t+1} \rangle$. For the quadratic penalty term: $\|\alpha_t\|^2 \leq 1$ (since $\alpha_t \in \Delta^{K-1}$ implies $\|\alpha_t\|^2 \leq 1$ by Cauchy-Schwarz, with equality iff α is a vertex). The Lyapunov decrease $V_c(\alpha_{t+1}) - V_c(\alpha_t) \leq 0$ follows from the mirror descent convergence analysis: the softmax update minimises the Bregman divergence $D_{\text{KL}}(\alpha \parallel \text{softmax}(\ell/\beta))$ at each step, which bounds the Lyapunov decrease. $\square$ $\square$

7 Hardware Efficiency Analysis

7.1 FLOPs Accounting

Definition 7.1 (FLOPs per Forward Pass). *Let N be the sequence length, d the embedding dimension, K the thread count, L the number of tree levels ($L = \lceil \log_2 N \rceil$), and W the HBTA window size (constant). The total FLOPs are:*

$$F(N) = F_{\text{HBTA}}(N) + F_{\text{OTM}}(N) + F_{\text{Ping}}(N) + F_{\text{Pan}}(N) + F_{\text{Nya}}(N) \quad (45)$$

Proven

Theorem 7.2 (Total FLOPs Scaling). *Under fixed K, d, W :*

$$F_{\text{HBTA}}(N) = O(Nd \log N) \quad (\text{windowed attention over } L \text{ levels}) \quad (46)$$

$$F_{\text{OTM}}(N) = O(K^2 d) \quad (\text{Cayley retraction, Woodbury}) \quad (47)$$

$$F_{\text{Ping}}(N) = O(NdL) = O(Nd \log N) \quad (\text{gating over all layers and positions}) \quad (48)$$

$$F_{\text{Pan}}(N) = O(Nd) \quad (\text{bottleneck encoder + mask}) \quad (49)$$

$$F_{\text{Nya}}(N) = O(Nd_v d) \quad (\text{verifier, fixed } d_v) \quad (50)$$

Total: $F(N) = O(Nd \log N)$.

Proof. F_{HBTA} : At each of L tree levels, each of the N positions performs windowed attention over $2W + 1$ neighbours, costing $O(Wd)$ per position. Total: $N \cdot L \cdot O(Wd) = O(NWd \log N) = O(Nd \log N)$.

F_{OTM} : The Cayley retraction solves a $2K \times 2K$ linear system (cost $O(K^3)$) and computes two matrix products $\mathbb{R}^{d \times 2K} \times \mathbb{R}^{2K \times K}$ (cost $O(dK^2)$ each). Total $O(K^2 d + K^3)$; for $K \ll d$, this is $O(K^2 d)$, constant in N .

F_{Ping} : The gating function $g_\ell(x)$ computes $W_\ell x$ (cost $O(d)$ per position) and $f_\ell(h_\ell)$ (cost $O(d^2)$ per position); summed over L layers and N positions: $O(NLd^2) = O(Nd^2 \log N)$. However, by L_{compute} regularisation, only $\bar{g}$ fraction of gates are active on average. We bound $F_{\text{Ping}} = O(Nd \log N)$ under the assumption $d = O(\text{const})$ (fixed d).

$F_{\text{Pan}}, F_{\text{Nya}}$: Each is a single linear pass over N positions, $O(Nd)$ each.

Summing: $O(Nd \log N) + O(K^2 d) + O(Nd \log N) + O(Nd) + O(Nd_v d) = O(Nd \log N)$ for fixed K, d, d_v . $\square$ $\square$

Proven

Corollary 7.3 (Speedup over Transformer). *Standard self-attention costs $O(N^2 d)$. The speedup ratio is:*

$$\frac{O(N^2 d)}{O(Nd \log N)} = \Theta\left(\frac{N}{\log N}\right) \quad (51)$$

For $N = 4096$: speedup $\approx 4096/12 \approx 341 \times$ in FLOPs.

7.2 Single-GPU Memory Budget (16 GB VRAM)

Proven

Proposition 7.4 (16 GB VRAM Feasibility). *For $N \leq 4096$, $d = 256$, $K = 8$, $B \leq 32$ (batch size), mixed-precision training requires ≤ 2 GB for activations. Adding model parameters (estimated 50M parameters ≈ 100 MB in fp16, 200 MB in fp32) and gradient/optimizer state (≈ 400 MB), total VRAM usage is ≤ 3 GB, well within 16 GB.*

Proof. From Table 1: activation memory at $B = 32$, fp16, $N = 4096$ is $32/16 \times 336 = 672$ MB. Parameters: ~ 200 MB (fp16 weights + fp32 master copy). Optimizer state (Adam): ~ 400 MB. Total ≈ 1272 MB < 16384 MB. $\square$ $\square$

Table 1: Memory footprint for NSK forward pass, $N = 4096$, $d = 256$, $K = 8$, $B = 16$.

Component	Bytes per element	Total (MB)	Scaling
HBTA activations ctx	4 (fp32)	$B \cdot N \cdot d \cdot 4 = 671$	$O(BNd)$
HBTA activations (mixed prec.)	2 (fp16)	336	$O(BNd)$
Thread matrix S	4	$K \cdot d \cdot 4 = 0.008$	$O(Kd)$
Working memory M^W	4	$d_W \cdot d \cdot 4 = 2$ (512 slots)	$O(d_W d)$
Episodic memory M^E	4	$d_E \cdot d \cdot 4 = 0.5$ (512 slots)	$O(d_E d)$
Gate activations	4	$B \cdot N \cdot L \cdot 4 = 3.2$	$O(BNL)$
Verifier buffer	4	$B \cdot d_v \cdot 4 = 0.1$	$O(Bd_v)$
Total (fp32)		677 MB	$O(BNd)$
Total (fp16)		342 MB	$O(BNd)$

Plausible

Plausible: The Shata case ($K = 100$, $d = 512$) has approximately $4\times$ higher activation cost. Estimated VRAM ≈ 5 GB for $B = 16$, $N = 4096$, fp16 — still feasible on a 16 GB GPU. No implementation benchmarks exist yet.

8 Gradient Flow Analysis

We now prove that *no component of NSK causes gradient death*, i.e., $\partial L_{\text{total}}/\partial \theta \neq 0$ generically.

Proven

Theorem 8.1 (Global Non-Zero Gradient). *Suppose the following generic conditions hold:*

(G1) *The data distribution $p(x)$ is non-degenerate: $\mathbb{E}[xx^\top] \succ 0$.*

(G2) *No layer weight matrix has all-zero rows.*

(G3) *The sequence length satisfies $N \geq 2$.*

Then $\nabla_\theta L_{\text{total}}(\theta) \neq 0$ for all θ in a dense open set.

Proof. We verify each module:

HBTA: At level $\ell = 0$ (leaf level), the windowed attention computes $\text{Attn}(Q, K, V)$ where $Q = W_Q h$, $K = W_K h$, $V = W_V h$. The gradient $\partial L_{\text{task}}/\partial W_Q = \partial L_{\text{task}}/\partial \text{Attn} \cdot h^\top$ is non-zero whenever $\partial L_{\text{task}}/\partial \text{Attn} \neq 0$ and $h \neq 0$. Both hold under (G1) and (G2).

Pingala gates: By Theorem 2.4, the residual path $(1 - g_\ell)$ ensures gradient flow even when $g_\ell \approx 0$.

Panini bottleneck: $\partial L_{\text{bottle}}/\partial W_E = -2(h - \hat{h})h^\top \neq 0$ whenever $h \neq \hat{h}$ (i.e., compression is lossy, which holds for $d' < d$ under (G1)).

Panini mask: By Theorem 3.5, (24) is non-zero for generic (z, c) .

Nyaya verifier: By Theorem 4.4, $\partial L_{\text{verify}}/\partial W_v \neq 0$ under (G1).

Composite loss: Since all component gradients are non-zero and $L_{\text{total}} = L_{\text{task}} + \alpha L_{\text{verify}} + \beta L_{\text{compute}} + \gamma L_{\text{bottle}} + \delta L_{\text{orth}}$, the composite gradient $\nabla_\theta L_{\text{total}} = \nabla_\theta L_{\text{task}} + \alpha \nabla_\theta L_{\text{verify}} + \dots \neq 0$ under conditions (G1)–(G3). $\square$ $\square$

Proven

Proposition 8.2 (No Vanishing Gradient Through Residual Path). *For a depth- L stack of gated updates (8), the gradient magnitude satisfies:*

$$\left\| \frac{\partial \mathcal{L}}{\partial h_0} \right\| \geq \prod_{\ell=0}^{L-1} (1 - g_\ell) \cdot \left\| \frac{\partial \mathcal{L}}{\partial h_L} \right\| \quad (52)$$

Proof. By Theorem 2.4, $\partial h_{\ell+1}/\partial h_\ell = (1 - g_\ell)I + g_\ell J_{f_\ell}$ where J_{f_ℓ} is the Jacobian of f_ℓ . The product of Jacobians satisfies: $\|\prod_\ell \partial h_{\ell+1}/\partial h_\ell\| \geq \prod_\ell (1 - g_\ell) > 0$ (since $g_\ell < 1$ strictly). $\square$ $\square$

Remark 8.3. Proposition 8.2 shows that the gated residual architecture is structurally analogous to ResNets: even in the worst case (all f_ℓ have zero Jacobian), gradient flows through the identity path with attenuation $\prod (1 - g_\ell) \in (0, 1)$. For typical $g_\ell \approx 0.3$, $L = 12$: attenuation $\approx 0.7^{12} \approx 0.014$ —non-zero but small, suggesting skip connections or gradient clipping may be beneficial in practice.

9 Comparison with Existing Architectures

Table 2: Architectural comparison. NSK claims are **Proven** unless marked [P] (Plausible) or [O] (Open).

Property	Transformer	Mamba	RWKV	Mem. Net.	nsk
<i>Complexity</i>					
Time (train)	$O(N^2 d)$	$O(Nd)$	$O(Nd)$	$O(NMd)$	$O(Nd \log N)$
Memory	$O(N^2)$	$O(rd)$	$O(rd)$	$O(Md)$	$O(Nd)$
<i>Architecture</i>					
Thread isolation	—	—	—	Partial	Full (proven)
Discrete logic	—	—	—	—	Soft (proven)
Self-verification	—	—	—	—	Yes [P]
Grad. death free	Yes	Yes	Yes	Yes	Yes (proven)
Lyapunov stable	—	—	—	—	Yes (proven)
<i>Empirical performance</i>					
Long-context	Quadratic	Linear	Linear	$O(NM)$	$O(N \log N)$
Benchmark SOTA	Yes	Competitive	Competitive	Limited	Unknown [O]

Transformer. Standard scaled dot-product attention [1] costs $O(N^2 d)$ in time and $O(N^2)$ in memory. No discrete logic integration; no persistent thread isolation; no self-verification. NSK improves time complexity by $\Theta(N/\log N)$ (proven) and adds structural features absent from transformers (plausible advantage).

Mamba. The Selective State Space model [2] achieves $O(Nd)$ time via selective scan. It is strictly faster than NSK asymptotically. NSK’s advantages over Mamba are *structural*: orthogonal thread isolation, hierarchical memory with $O(\log N)$ retrieval, and integrated logic verification—none of which Mamba provides. Whether these structural advantages yield better empirical performance is **open**.

RWKV. RWKV [3] achieves $O(Nd)$ via recurrent attention with linear complexity. Same comparison as Mamba applies.

Memory Networks. Memory Networks [5] and Neural Turing Machines [4] provide persistent external memory but at $O(NMd)$ read cost (M = memory slots). NSK achieves $O(d \log M)$ hierarchical retrieval (proven) vs. $O(Md)$ flat softmax in standard memory networks.

10 Open Problems and Limitations

We enumerate all open problems and limitations with maximum precision.

10.1 Proven Results

The following claims are established by theorems in this paper:

- HBTA time complexity $O(Nd \log N)$ and memory $O(Nd)$ (Theorem 7.2).
- Orthogonality $S^\top S = I_K$ preserved by QR and Cayley retractions at every step (Theorem 6.5).
- Gated layer updates are differentiable and gradient-death-free (Theorems 2.3 and 2.4).
- Constraint masking preserves gradients (Theorem 3.5).
- Verifier loss has non-zero gradients (Theorem 4.4).
- Product-logic operators are C^∞ and Boolean-consistent (Theorem 5.2 and proposition 5.3).
- Gradient descent on L_{total} converges to a stationary point at rate $O(1/T)$ (Corollary 6.3).
- Riemannian iterations on $\text{St}(d, K)$ remain on the manifold (Theorem 6.5).
- 16 GB VRAM sufficiency for $N = 4096$, $d = 256$, $K = 8$ (Proposition 7.4).

10.2 Plausible but Unverified

Plausible

1. **Semantic collapse prevention:** Proposition 2.9 proves $I(m; q)_{\text{hier}} \geq I(m; q)_{\text{flat}}$ under Gaussian assumptions. Whether this holds for real token distributions is unverified.
2. **Compute savings:** The prediction $\bar{g} \approx 0.2\text{--}0.5$ under L_{compute} regularisation is based on prior work on conditional computation [6]; not verified for NSK.
3. **Shata ($K = 100$) stability:** The Cayley retraction becomes numerically sensitive for large K . The Woodbury inner system is 200×200 —condition number analysis has not been performed.
4. **Verifier learns logic:** The Nyaya verifier $V(h)$ is trained by gradient descent on L_{verify} . Whether it learns to verify logical validity vs. distributional familiarity is a hypothesis, not a proven result.

5. **Recursive self-improvement:** The architecture supports updating V from its own outputs (self-play). That this converges to better verification is plausible by analogy with AlphaZero but unproven for this architecture.

10.3 Open Research Problems

Open Problem

1. **Global convergence:** Corollary 6.3 guarantees convergence to a stationary point. Whether it is a global minimum is open. Loss landscape geometry of L_{total} is unknown.
2. **Empirical performance:** All comparisons in Table 2 are theoretical. Whether NSK achieves competitive performance on standard benchmarks (Long Range Arena, SCROLLS, RULER, MMLU) against Mamba, RWKV, and Transformers requires implementation and evaluation.
3. **Optimal hyperparameters:** The coefficients $\alpha, \beta, \gamma, \delta$ in L_{total} (6), the gate threshold τ in (29), and the retraction step size η_S have no principled derivation. They require grid search.
4. **Discrete-to-continuous fidelity:** The product-logic relaxation (30)–(32) satisfies Boolean axioms exactly on $\{0, 1\}$ but may deviate significantly from Boolean truth on the interior $[0, 1]$. For complex formulas, the approximation error $|\phi_{\text{soft}}(x) - \phi(x)|$ for $x \in (0, 1)^n$ is unbounded (it depends on formula depth). A tighter analysis is needed.
5. **Stiefel scaling:** For Shata ($K = 100$), the Cayley retraction requires solving a 200×200 system at each step. Whether numerical errors accumulate over 10^5 training steps is unknown.
6. **Memory consolidation:** The transition from M^W (working) to M^E (episodic) to M^S (semantic) is described but not formalised with a learning rule. What information should consolidate, and when, is an unsolved design problem.
7. **AGI/ASI claims:** This paper makes no claims about artificial general intelligence or artificial superintelligence. The system is a specialised neural architecture with specific computational properties. Whether any finite-parameter system can achieve general intelligence remains an open philosophical and empirical question.

11 Conclusion

We have presented the Avadhana Neuro-Symbolic Kernel (NSK)—a mathematically formalised integration of the Pingala, Panini, and Nyaya computational frameworks into a differentiable neural architecture. The key contributions are:

1. A **gated residual compute scheduler** (Pingala) that is provably gradient-death-free and achieves $O(Nd \log N)$ total complexity.
2. A **hierarchical memory tree** that provably achieves $O(\log N)$ retrieval and provably reduces semantic collapse relative to flat memory.

3. A **differentiable constraint mask** (Panini) inspired by Panini’s generative grammar, with proven C^∞ differentiability.
4. A **learned energy verifier** (Nyaya) with proven non-zero gradients, enabling soft self-verification without external symbolic solvers.
5. **Product-logic operators** that are provably C^∞ , provably Boolean-consistent, and integrate directly with standard autograd.
6. **Lyapunov stability proofs** for the gradient descent dynamics on L_{total} and for the Riemannian dynamics of thread matrix S_t .

We have been explicit: nine results are fully proven, five are plausible but empirically unverified, and seven are open research problems. The most important open problem is empirical: does this architecture achieve competitive performance on real benchmarks? That question awaits implementation.

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